Direct-Current Circuits

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Learning Outcomes

In this chapter, you'll learn…

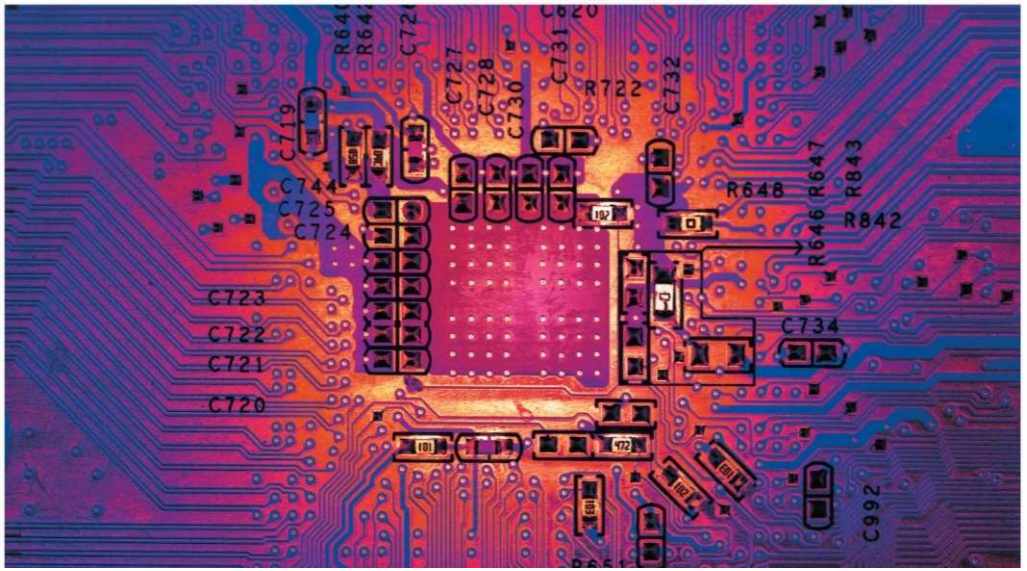
- how to analyze circuits with multiple resistors in series or parallel.
- rules that you can apply to any circuit with more than one loop.
- how to use an ammeter, voltmeter, ohmmeter, or potentiometer in a circuit.
- how to analyze circuits that include both a resistor and a capacitor.
- how electric power is distributed in the home.

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Introduction

- Even in a complex circuit like the one on this circuit board, several resistors with different resistances can be connected so that all of them have the same potential difference; in this case the currents through the resistors will be different.
- In this chapter, we will learn general methods for analyzing complex networks of resistors, batteries, and capacitors.
- We shall look at various instruments for measuring electrical quantities in circuits.



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DC Versus AC

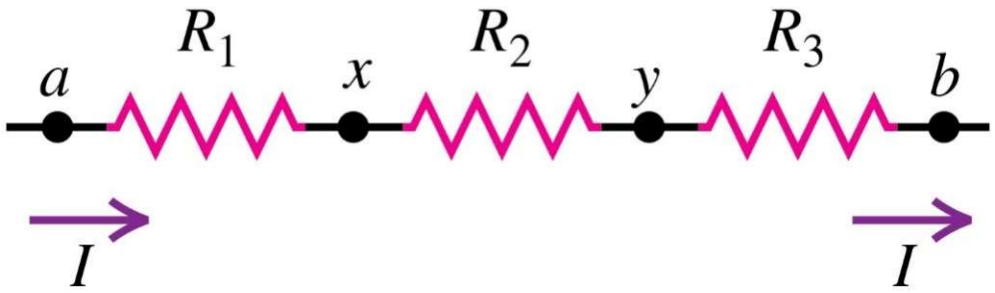
- Our principal concern in this chapter is with **direct-current** (dc) circuits, in which the direction of the current does not change with time.
- Flashlights and automobile wiring systems are examples of direct-current circuits.
- Household electrical power is supplied in the form of **alternating current** (ac), in which the current oscillates back and forth.
- The same principles for analyzing networks apply to both kinds of circuits, and we conclude this chapter with a look at household wiring systems.

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Resistors in Series

- Resistors are in **series** if they are connected one after the other so the current is the same in all of them.



- The **equivalent resistance** of a series combination is the **sum** of the individual resistances:

Resistors in series:

$$R_{eq} = R_1 + R_2 + R_3 + \cdots$$

Equivalent resistance of series combination

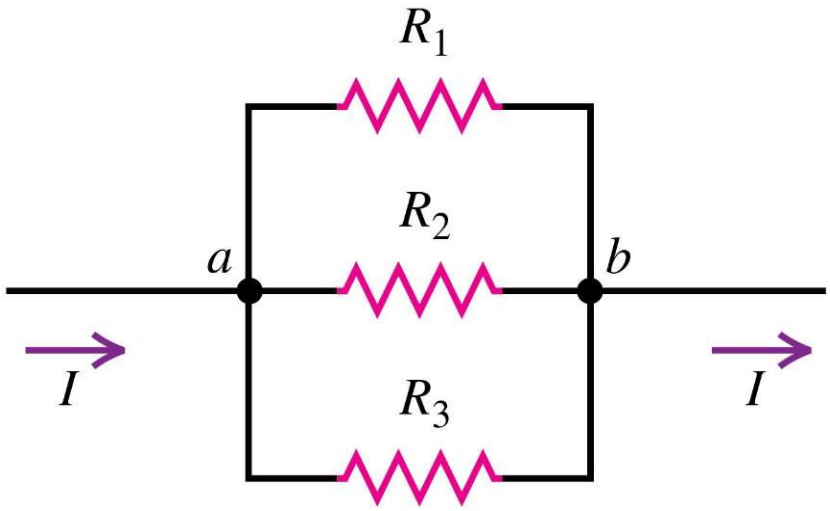
Resistances of individual resistors

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Resistors in Parallel

- If the resistors are in **parallel**, the current through each resistor need not be the same, but the potential difference between the terminals of each resistor must be the same, and equal to V_{ab}.



- The reciprocal of the equivalent resistance of a parallel combination equals the sum of the reciprocals of the individual resistances:

Resistors in parallel:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots$$

Equivalent resistance of parallel combination

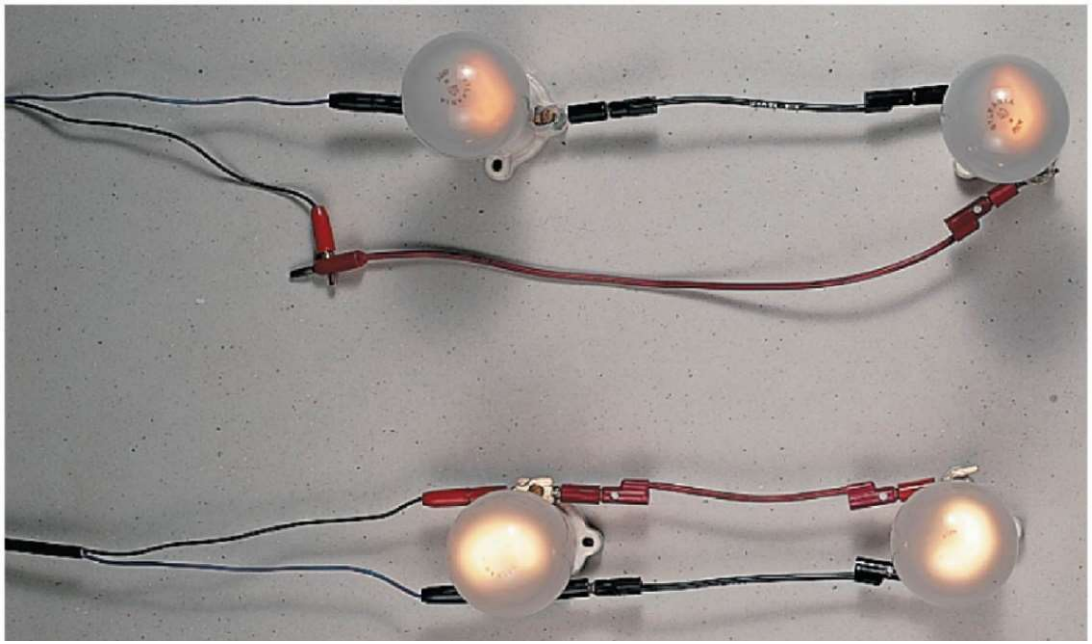
Resistances of individual resistors

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Series versus Parallel Combinations

- When connected to the same source, two incandescent light bulbs in series (shown at top) draw less power and glow less brightly than when they are in parallel (shown at bottom).



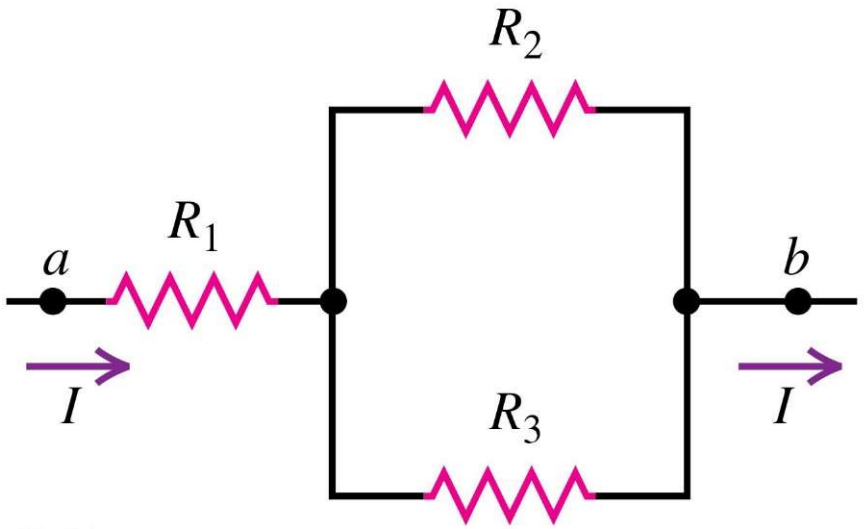
- Video Tutor Demonstration: Bulbs Connected in Series and Parallel

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Series and Parallel Combinations: Example 1

- Resistors can be connected in **combinations** of series and parallel, as shown.
- In this case, try reducing the circuit to series and parallel combinations.



- For the example shown, we first replace the parallel combination of R_2 and R_3 with its equivalent resistance; this then forms a series combination with R_1.

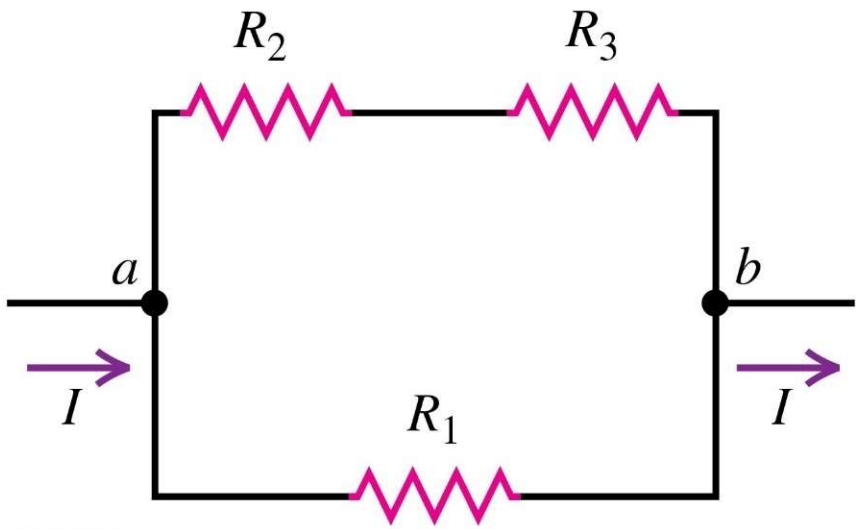
- Video Tutor Solution: Example 26.1

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Series and Parallel Combinations: Example 2

- Resistors can be connected in **combinations** of series and parallel, as shown.
- In this case, try reducing the circuit to series and parallel combinations.



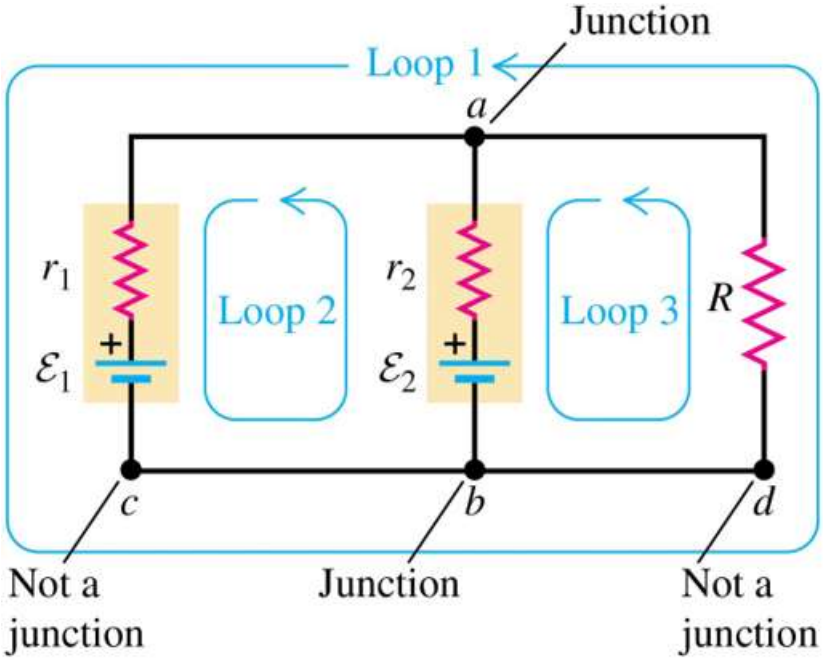
- For the example shown, we first replace the series combination of R_2 and R_3 with its equivalent resistance; this then forms a parallel combination with R_1.

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Kirchhoff's Rules

- Many practical resistor networks cannot be reduced to simple series-parallel combinations.
- To analyze these networks, we'll use the techniques developed by Kirchhoff.



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Kirchhoff's Junction Rule

- A **junction** is a point where three or more conductors meet.

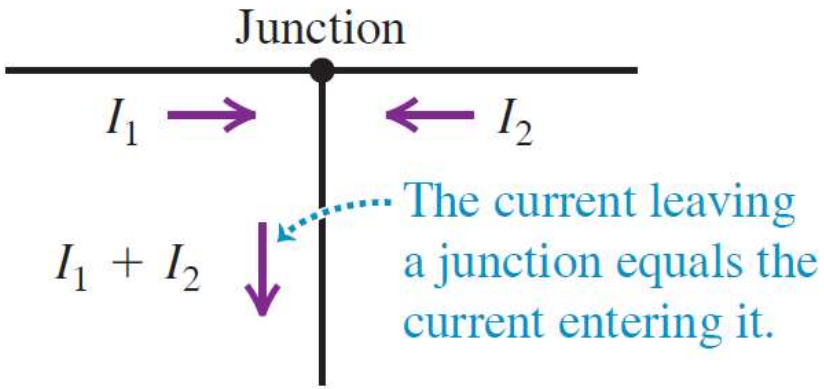
Kirchhoff's junction rule
(valid at any junction):

The sum of the currents into any junction ...

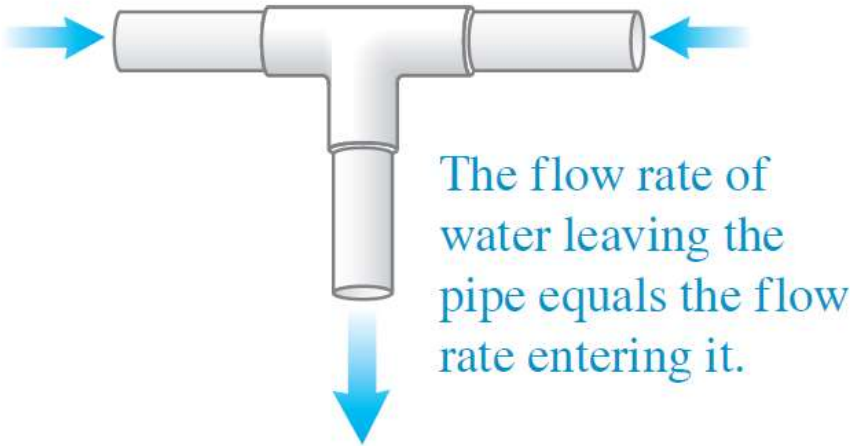
$$\sum I = 0$$

... equals zero.

(a) Kirchhoff's junction rule



(b) Water-pipe analogy



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Kirchhoff's Loop Rule

- A **loop** is any closed conducting path.
- Kirchhoff's loop rule (valid for any closed loop) is:

Kirchhoff's loop rule (valid for any closed loop): The sum of the potential differences around any loop ... $\sum V = 0$ ... equals zero.

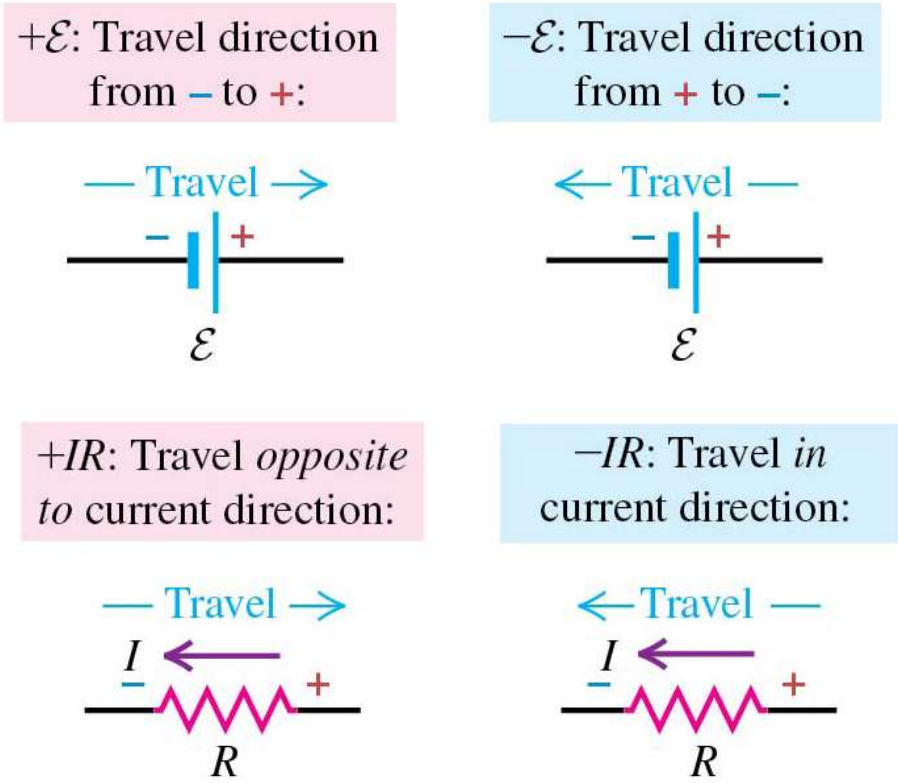
- The loop rule is a statement that the electrostatic force is **conservative**.

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Sign Conventions for the Loop Rule

- Use these sign conventions when you apply Kirchhoff's loop rule.



- In each part of the figure, "Travel" is the direction that we imagine going around the loop, which is not necessarily the direction of the current.

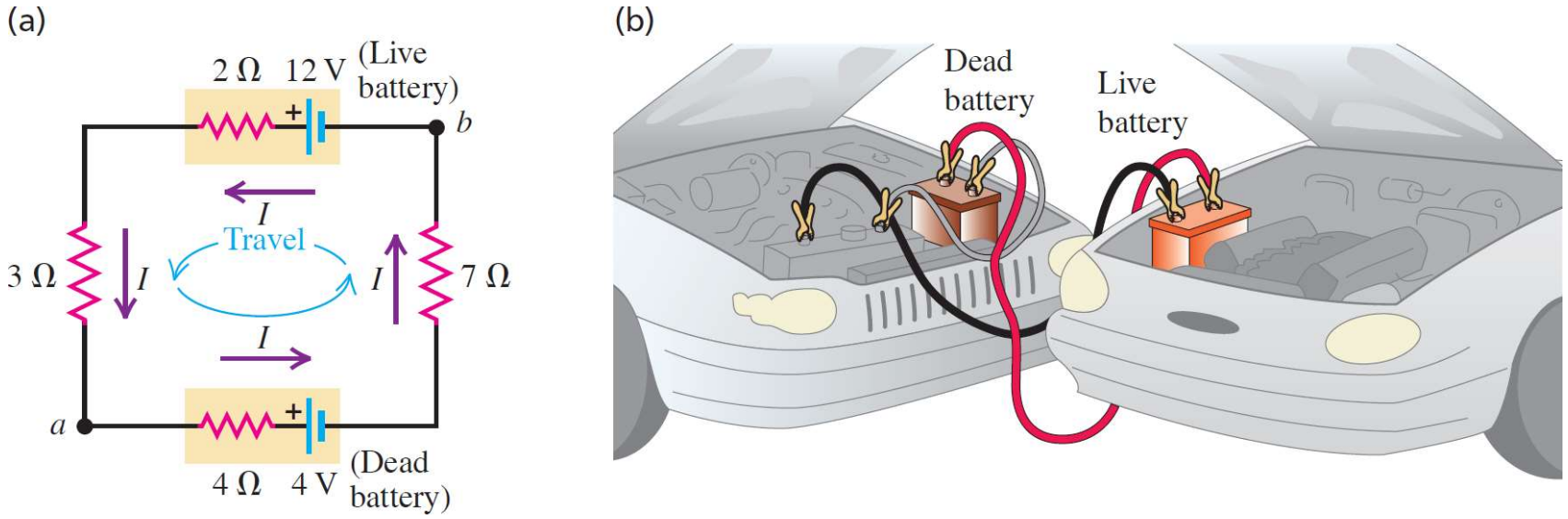
- Video Tutor Solution: Example 26.6

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A Single-Loop Circuit

- The circuit shown contains two batteries, each with an emf and an internal resistance, and two resistors.
- Using Kirchhoff's rules, you can find the current in the circuit, the potential difference V_{ab}, and the power output of the emf of each battery.

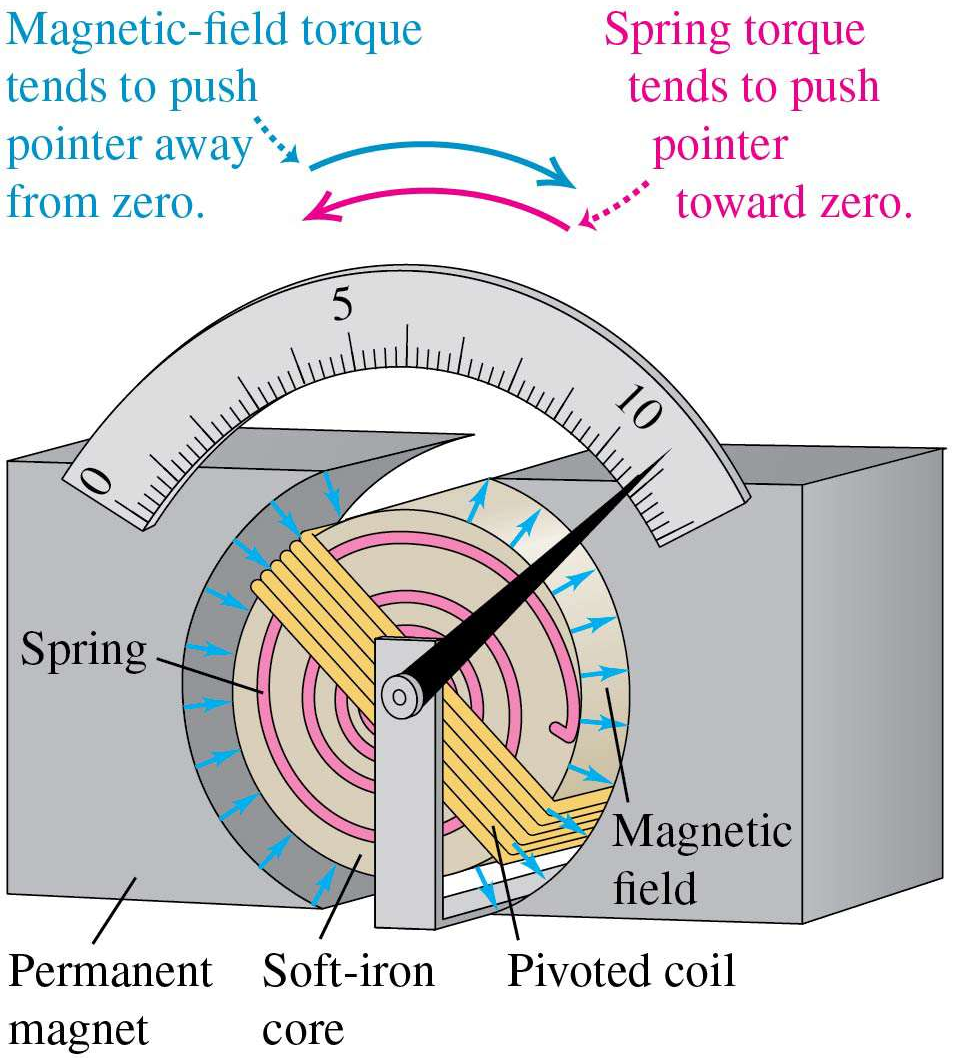


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D'Arsonval Galvanometer

- A **galvanometer** measures the current that passes through it.
- Many electrical instruments, such as ammeters and voltmeters, use a galvanometer in their design.

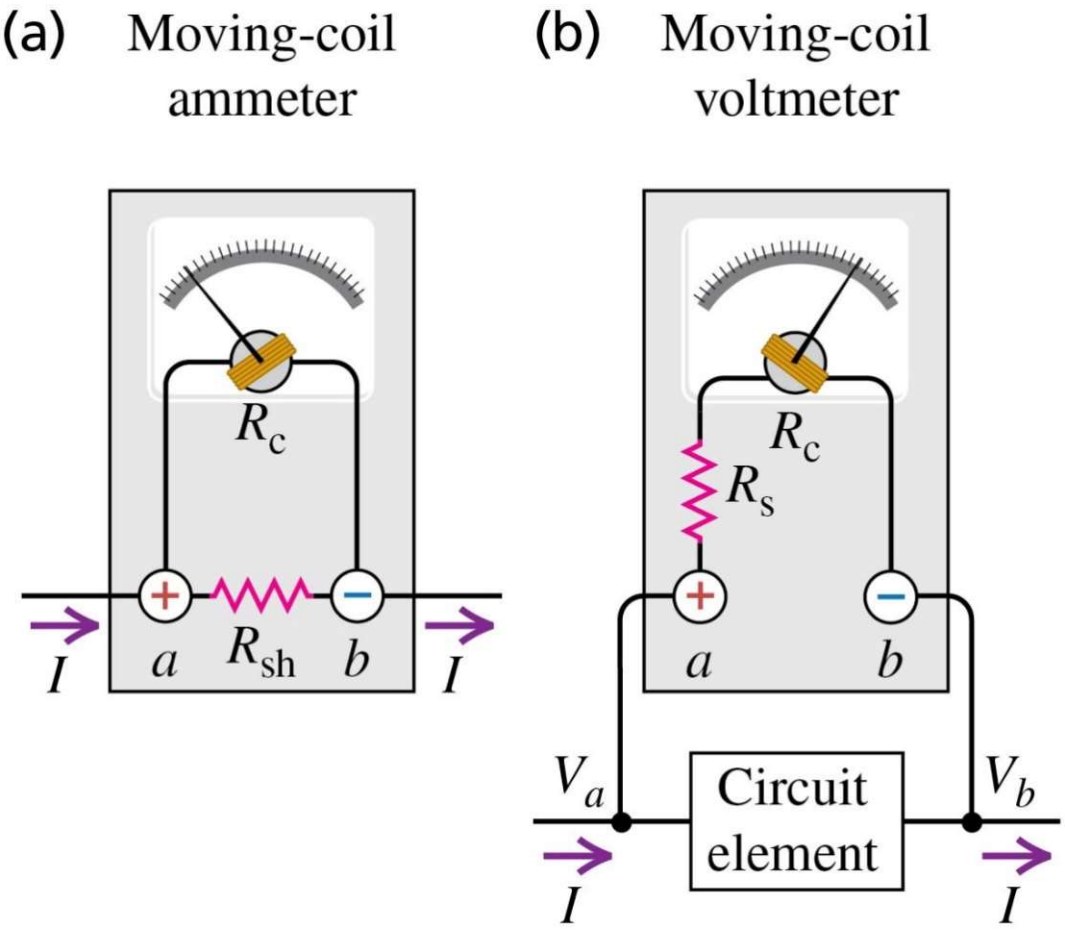


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Ammeters and Voltmeters

- An **ammeter** measures the current passing through it.
- A **voltmeter** measures the potential difference between two points.
- Both instruments contain a galvanometer.

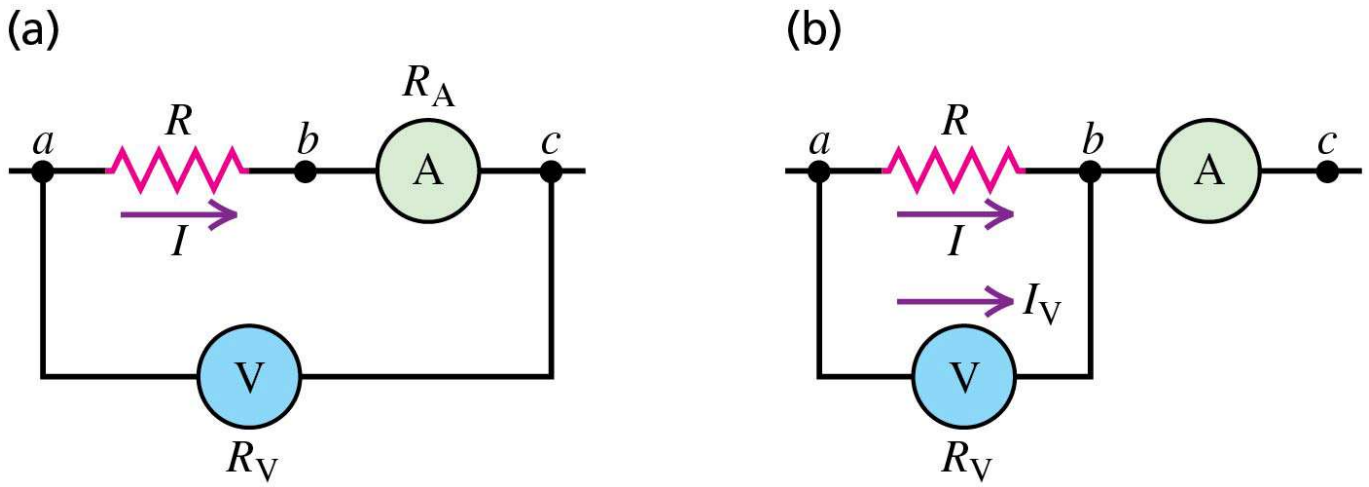


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Ammeters and Voltmeters in Combination

- An ammeter and a voltmeter may be used together to measure resistance and power.
- Two ways to do this are shown below.
- Either way, we have to correct the reading of one instrument or the other unless the corrections are small enough to be negligible.



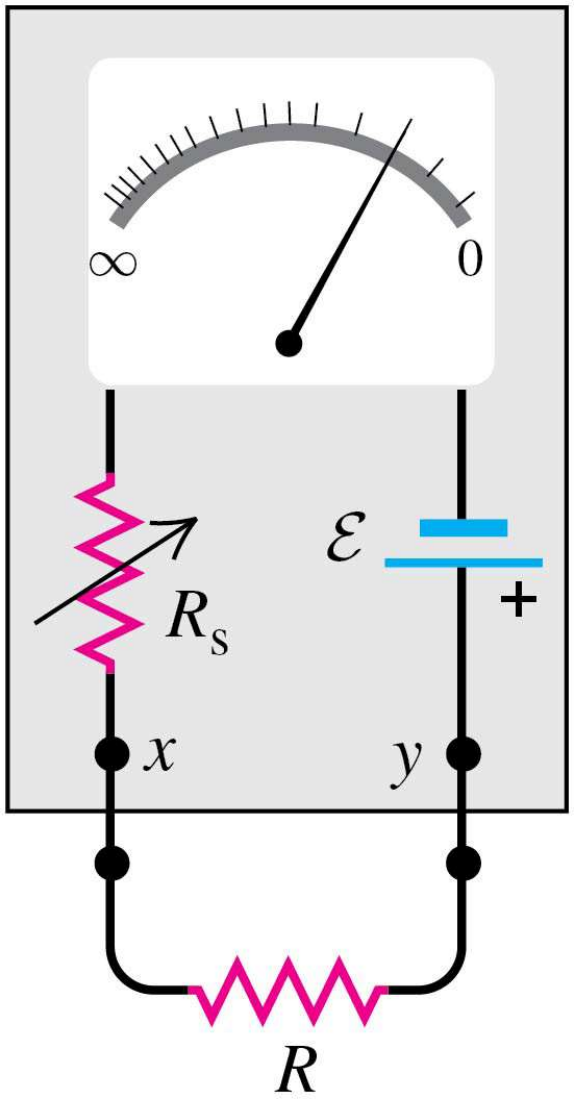
- Video Tutor Solution: Example 26.10

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Ohmmeters

- An **ohmmeter** consists of a meter, a resistor, and a source (often a flashlight battery) connected in series.
- The resistor R_s has a variable resistance, as is indicated by the arrow through the resistor symbol.
- To use the ohmmeter, first connect *x* directly to *y* and adjust R_s until the meter reads zero.
- Then connect *x* and *y* across the resistor *R* and read the scale.



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Digital Multimeters

- A digital multimeter can measure voltage, current, or resistance over a wide range.

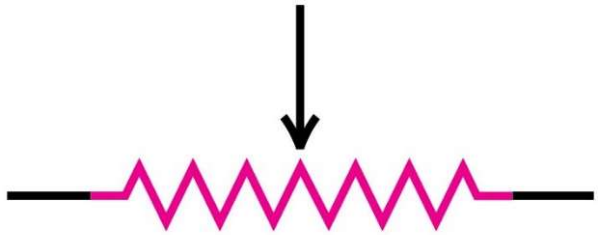


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The Potentiometer

- The **potentiometer** is an instrument that can be used to measure the emf of a source without drawing any current from the source.
- Essentially, it balances an unknown potential difference against an adjustable, measurable potential difference.
- The term potentiometer is also used for any variable resistor, usually having a circular resistance element and a sliding contact controlled by a rotating shaft and knob.
- The circuit symbol for a potentiometer is shown to the right.

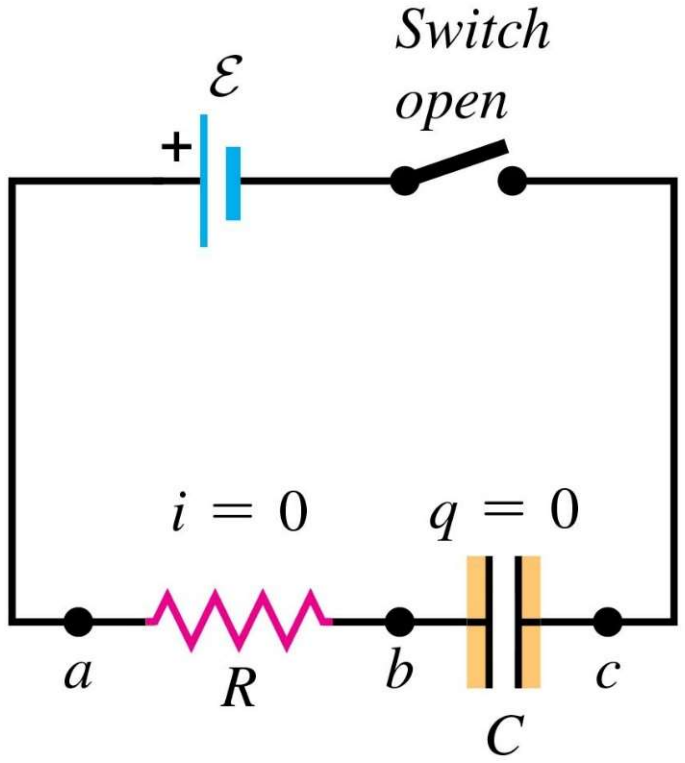


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R-C Circuits: Charging a Capacitor (1 of 4)

- Shown is a simple *R-C* circuit for charging a capacitor.
- We idealize the battery to have a constant emf and zero internal resistance, and we ignore the resistance of all the connecting conductors.
- We begin with the capacitor initially uncharged.

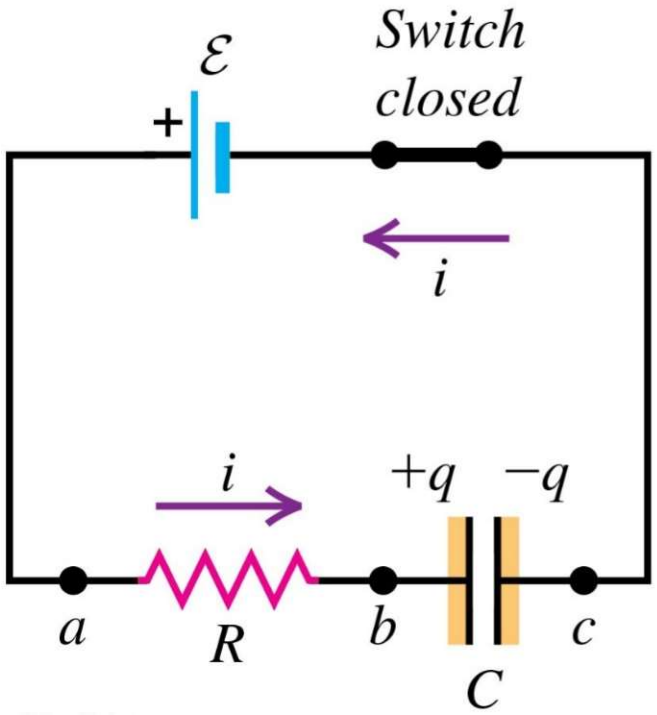


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R-C Circuits: Charging a Capacitor (2 of 4)

- At some initial time $t = 0$ we close the switch, completing the circuit and permitting current around the loop to begin charging the capacitor.
- As t increases, the charge on the capacitor increases, while the current decreases.



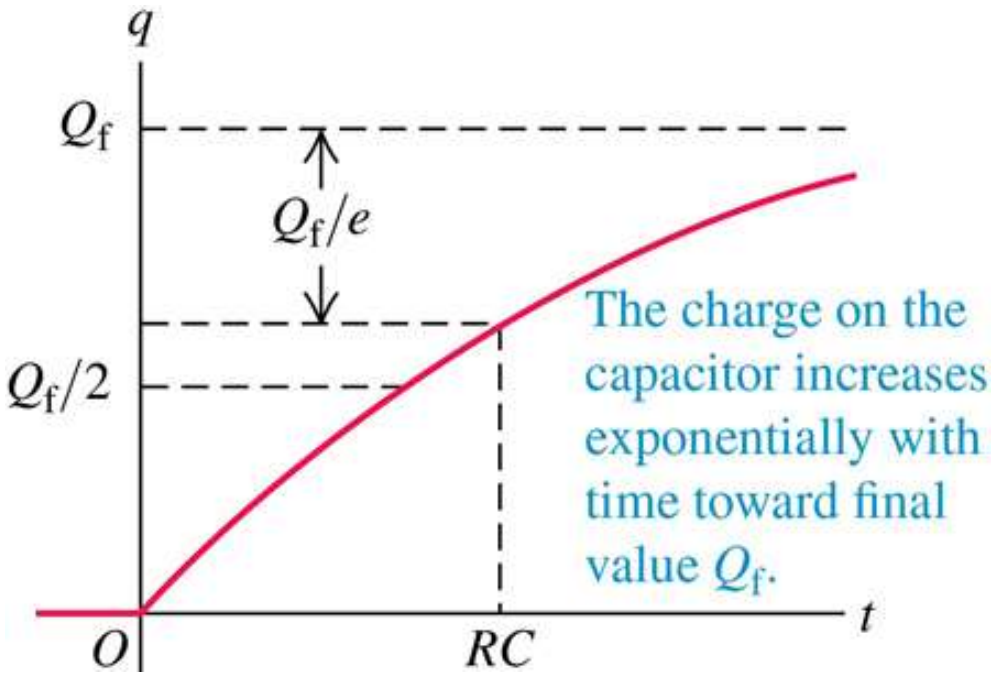
When the switch is closed, the charge on the capacitor increases over time while the current decreases.

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R-C Circuits: Charging a Capacitor (3 of 4)

- The charge on the capacitor in a charging *R-C* circuit increases exponentially, with a time constant $\tau = RC$.



R-C circuit, charging capacitor:

$$q = C\mathcal{E}(1 - e^{-t/RC}) = Q_f(1 - e^{-t/RC})$$

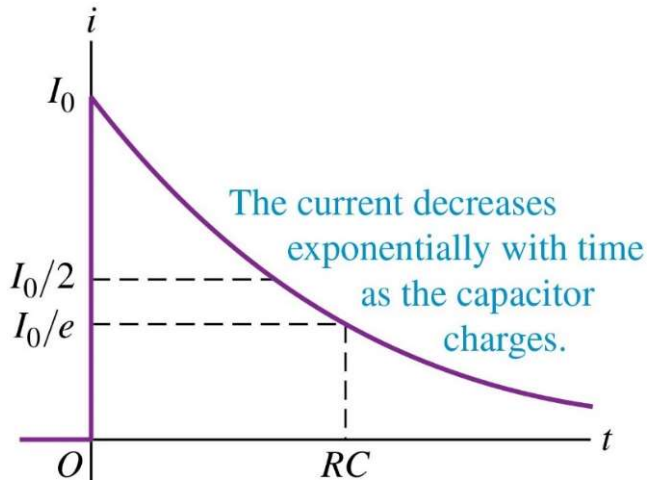
Capacitor charge; Capacitance; Final capacitor charge $= C\mathcal{E}$; Battery emf; Time since switch closed; Resistance

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R-C Circuits: Charging a Capacitor (4 of 4)

- The current through the resistor in a charging *R-C* circuit decreases exponentially, with a time constant $\tau = RC$.



R-C circuit, charging capacitor:

$$i = \frac{dq}{dt} = \frac{\mathcal{E}}{R}e^{-t/RC} = I_0 e^{-t/RC}$$

Current; Battery emf; Time since switch closed; Initial current $= \mathcal{E}/R$; Rate of change of capacitor charge; Resistance; Capacitance

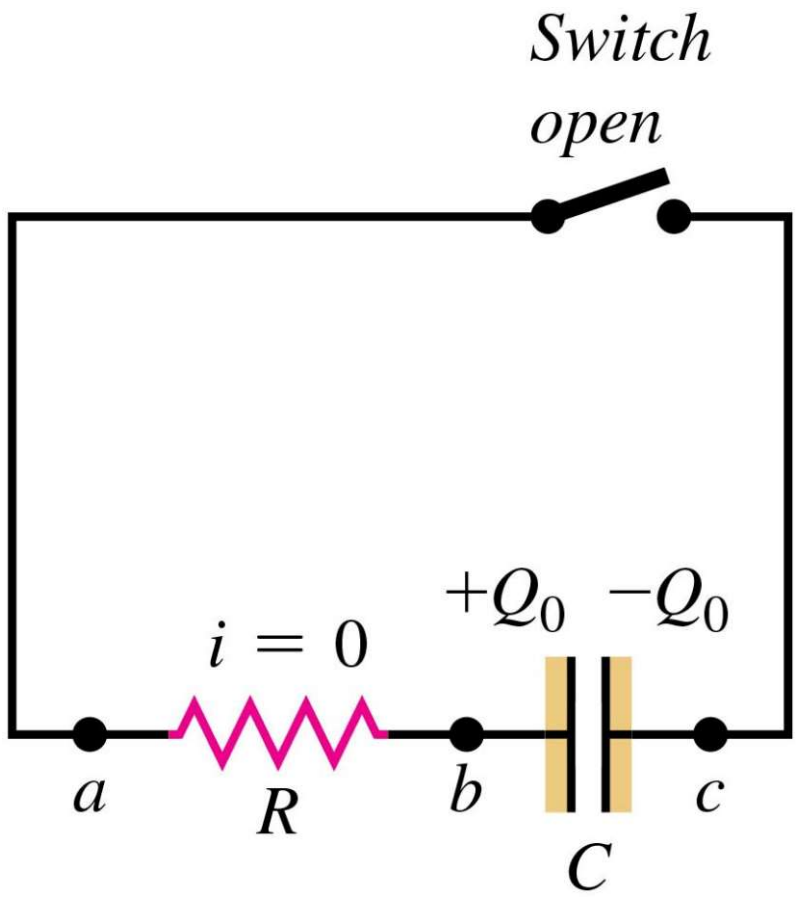
- Video Tutor Solution: Example 26.12

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R-C Circuits: Discharging a Capacitor (1 of 4)

- Shown is a simple *R-C* circuit for discharging a capacitor.
- Before the switch is closed, the capacitor charge is Q_0, and the current is zero.

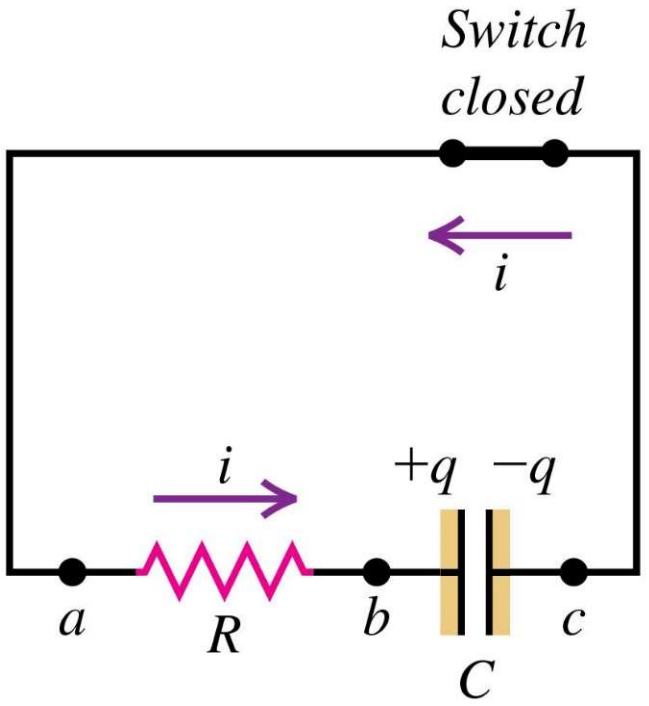


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R-C Circuits: Discharging a Capacitor (2 of 4)

- At some initial time $t = 0$ we close the switch, allowing the capacitor to discharge through the resistor.
- As t increases, the magnitude of the current decreases, while the charge on the capacitor also decreases.



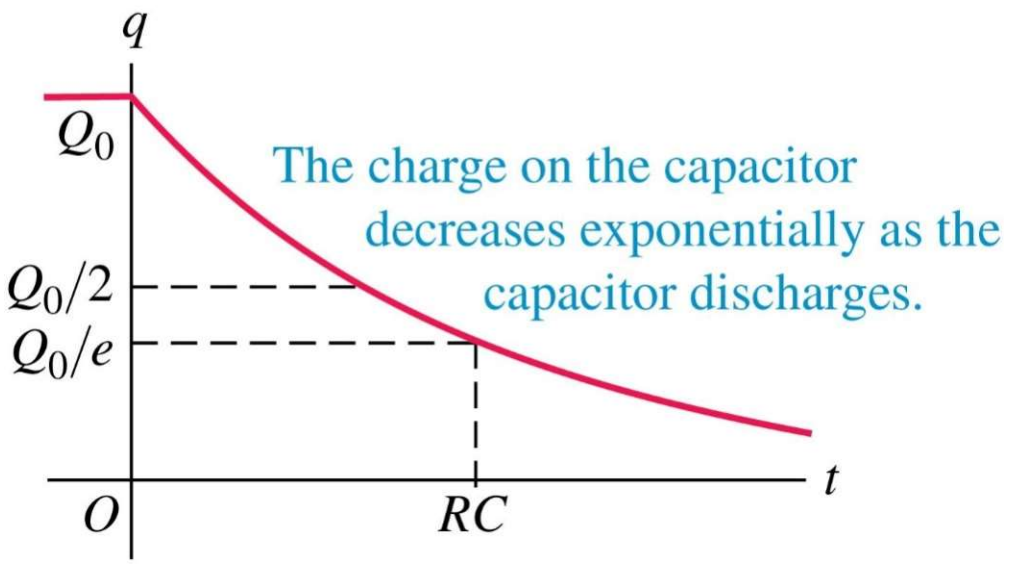
When the switch is closed, the charge on the capacitor and the current both decrease over time.

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R-C Circuits: Discharging a Capacitor (3 of 4)

- The charge on the capacitor in a discharging *R-C* circuit decreases exponentially, with a time constant $\tau = RC$.



R-C circuit, discharging capacitor:

$$q = Q_0 e^{-t/RC}$$

Capacitor charge

Initial capacitor charge

Capacitance

Time since switch closed

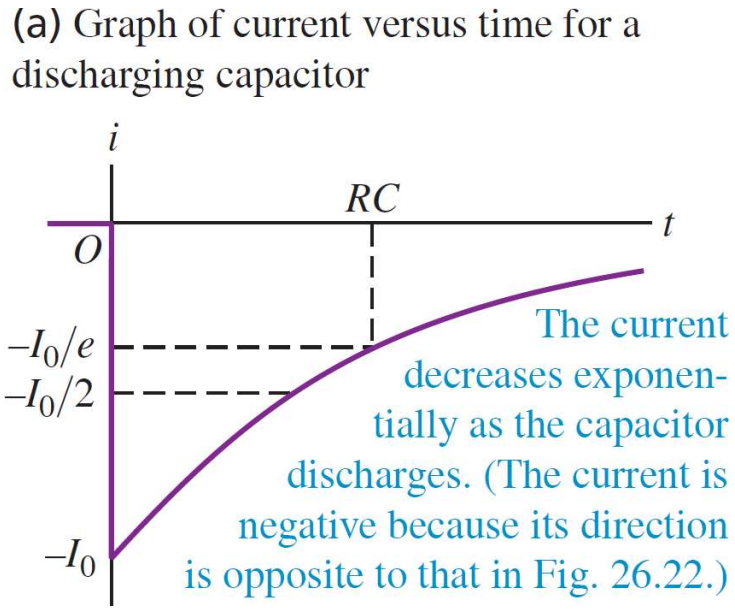
Resistance

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R-C Circuits: Discharging a Capacitor (4 of 4)

- The magnitude of the current through the resistor in a discharging *R-C* circuit decreases exponentially, with a time constant $\tau = RC$.



R-C circuit, discharging capacitor:

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC}e^{-t/RC} = I_0 e^{-t/RC}$$

Current; Rate of change of capacitor charge; Initial capacitor charge; Capacitance; Resistance; Time since switch closed; Initial current $= -Q_0/RC$

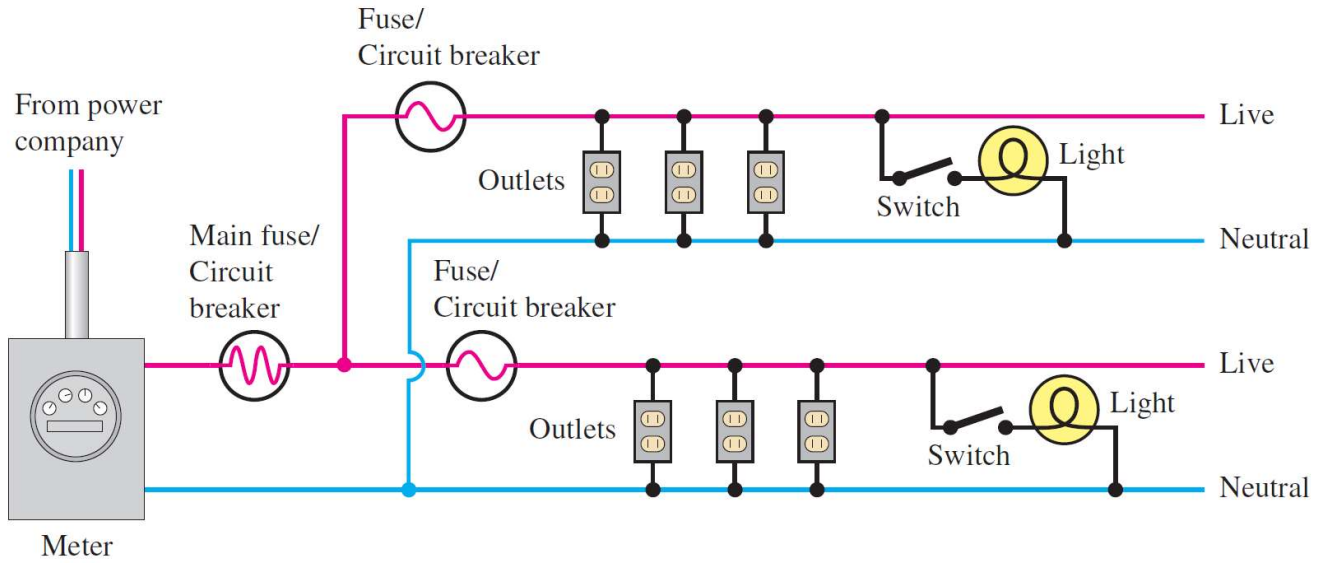
- Video Tutor Solution: Example 26.13

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Power Distribution Systems

- The figure below shows the basic idea of house wiring.
- The “live” has an alternating sinusoidal voltage with a root-mean-square value of 220–240 V in most of Europe and 120 V in USA.
- The “neutral” is connected to “earth,” which is usually an electrode driven into the earth.



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Circuit Overloads

- A **fuse** (Figure a) contains a link of lead–tin alloy with a very low melting temperature; the link melts and breaks the circuit when its rated current is exceeded.
- A **circuit breaker** (Figure b) is an electromechanical device that performs the same function, using an electromagnet or a bimetallic strip to "trip" the breaker and interrupt the circuit when the current exceeds a specified value.
- Circuit breakers have the advantage that they can be reset after they are tripped, while a blown fuse must be replaced.

(a)



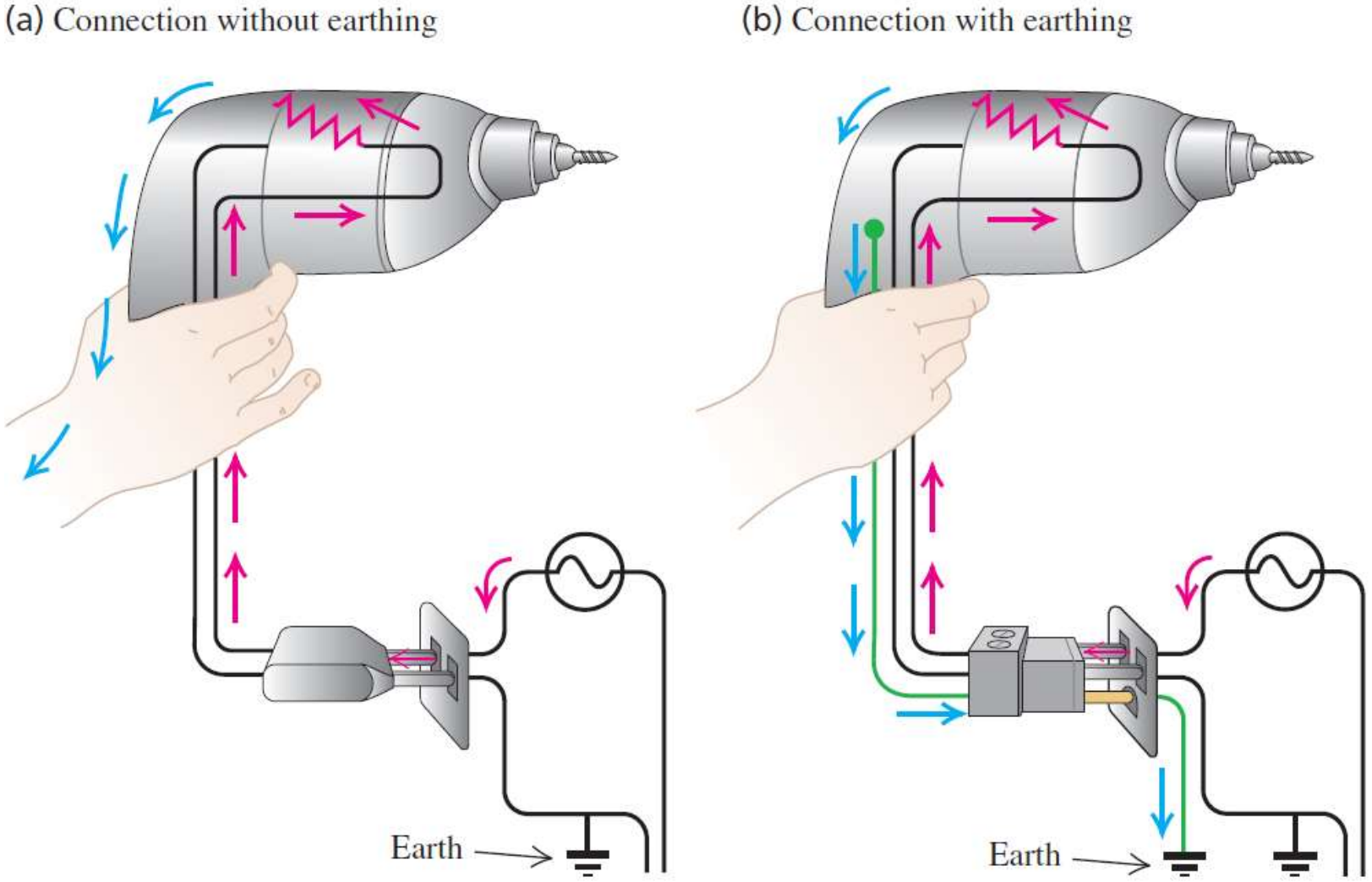
(b)



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Why It Is Safer to Use a Connection with Earthing



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